

Theoretical Bounds and Mathematical Mechanisms for Neural Scaling Laws:

A Cross-Disciplinary Mathematical Survey

Theoretical Machine Learning & Information Theory Synthesis

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Abstract

Empirical scaling laws in deep learning demonstrate that held-out cross-entropy loss decreases predictably as a power law of model parameters N , dataset tokens D , and compute budget $C \approx 6ND$, following the parametric form $L(N, D) = E + AN^{-\alpha_N} + BD^{-\alpha_D}$. Over the past several years, theoretical research across Information Theory, Differential Geometry, Statistical Physics, and Approximation Theory has revealed the exact mathematical mechanisms that give rise to these power laws.

This monograph provides a self-contained, rigorous, and pedagogically accessible synthesis of these theoretical foundations. We systematically review: **(i) Information Theory and Rate-Distortion Theory**, clarifying why neural scaling reflects lossy compression of infinite-dimensional heavy-tailed sources ($D(R) \sim R^{-\rho}$) rather than classical channel coding error exponents; **(ii) Language Statistics and Horizon-Limited Scaling**, examining Cagnetta et al.’s derivation of the data exponent $\alpha_D = \gamma/(2\beta)$ from token correlations and conditional entropy decay; **(iii) High-Dimensional Geometry and Approximation Bounds**, detailing Sharma–Kaplan manifold tiling ($N^{-4/d}$), Bahri et al.’s four-regime phase diagram, and DeVore–Howard–Micchelli n -width lower bounds; **(iv) Statistical Physics and Kernel Dynamics**, presenting exact dual scaling laws for linear models $\Theta(N^{-(b-1)} + D^{-(b-1)/a})$ and Neural Tangent Kernel spectra; and **(v) Combinatorial Quanta and Superposition**, analyzing Zipf-distributed skill quantization and interference noise scaling.

To bridge diverse disciplines, we provide an intuitive “Rosetta Stone” translation across mathematical dialects, step-by-step proofs of foundational theorems, and an explicit analysis of the “honest gap” between theoretical parameterizations and empirical observation.

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1 Scope, Terminology, and Standards of Evidence

1.1 The Empirical Phenomenon and Empirical Fits

Let N denote model capacity (typically the non-embedding parameter count), D the dataset size (the number of processed training tokens), and $C \approx 6ND$ the floating-point operations (FLOPs) expended during autoregressive transformer pretraining. A ubiquitous empirical finding in deep learning is that validation loss $L(N, D)$ decays smoothly over multiple orders of magnitude according to a joint power law:

$$L(N, D) = E + \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{D_c}{D}\right)^{\alpha_D} = E + AN^{-\alpha_N} + BD^{-\alpha_D}, \quad (1)$$

where E represents the irreducible entropy floor (the Bayes risk of the underlying data distribution), $N_c, D_c, A, B > 0$ are fit constants, and $\alpha_N, \alpha_D > 0$ are characteristic scaling exponents.

Equation (1) was documented across multiple modalities by Hestness et al. [17] and Rosenfeld et al. [35], and popularized for large language models by Kaplan et al. [24] and Hoffmann et al. (Chinchilla) [19]. In standard English autoregressive language modeling, typical empirical values fall in the ranges:

$$\alpha_N \approx 0.08 \text{ to } 0.34, \quad \alpha_D \approx 0.10 \text{ to } 0.28, \quad E \approx 1.5 \text{ to } 2.1 \text{ nats/token.} \quad (2)$$

1.2 A Cross-Disciplinary Rosetta Stone

A primary challenge in studying neural scaling laws is that researchers in Information Theory, Statistical Physics, Differential Geometry, and Deep Learning use different mathematical languages to describe identical underlying phenomena. Table 1 establishes a cross-disciplinary concordance.

Concept	Information Theory	Geometry / Approximation	Statistical Physics / ML
Data Complexity	Covariance spectrum $\lambda_k \sim k^{-(1+\rho)}$; Metric entropy $H(\varepsilon) \sim \varepsilon^{-d/s}$	Manifold intrinsic dimension d ; Sobolev smoothness s	Kernel / NTK spectrum $\lambda_k \sim k^{-a}$; Target power $c_k^2 \sim k^{-b}$
Model Capacity	Description rate $R \approx Nb$ bits; Representation states	Tiling cells $K \asymp N$; Linear regions; Metric covering	Parameter count N ; Feature width w ; Mode cutoff k_*
Data Budget	Sequence blocklength n ; Sample count D	Metric sample covering radius $\varepsilon \sim D^{-1/d}$	Training samples / tokens D ; Gradient update steps
Limiting Loss	Distortion-rate function $D(R) \sim R^{-\rho}$; Entropy rate h	Piecewise Taylor error $N^{-4/d}$; Nonparametric rate $D^{-\frac{2s}{2s+d}}$	Asymptotic generalization error $\mathcal{E}(D) \sim D^{-\frac{a+b-1}{a}}$; Bayes risk E

Table 1: Cross-disciplinary ‘‘Rosetta Stone’’ translating scaling concepts across mathematical fields.

1.3 Hierarchy of Scientific Evidence

Throughout this monograph, we classify theoretical statements according to four explicit levels of epistemic certainty:

1. **Rigorous Mathematical Theorem:** A closed-form theorem with complete proof for a fully specified mathematical source, function space, loss metric, and algorithm (e.g., Gaussian rate-distortion reverse water-filling, DeVore–Howard–Micchelli n -width lower bounds, Yang–Barron minimax rates).
2. **Controlled Solvable Model:** An exact analytic or asymptotic result derived in a stylized model (e.g., random feature models, high-dimensional ridge regression, infinite-width Neural Tangent Kernel limits, associative memory networks).
3. **Mechanistic Proposal:** A proposed physical or geometric mechanism transferring insights from a solvable model to finite, trained neural networks, accompanied by testable, falsifiable assumptions (e.g., Voronoi manifold tiling, Zipfian quantization of skills, token-correlation horizon cutoff).
4. **Empirical Observation:** Parametric curves fitted to observed training runs over a finite resource window.

2 What Empirical Scaling Laws Establish in Practice

2.1 The Chinchilla Compute-Optimal Allocation Rule

Given a total training compute budget $C \approx 6ND$ FLOPs, how should compute be optimally partitioned between parameter count N and token count D ? Minimizing the empirical loss ansatz $L(N, D) = E + AN^{-\alpha_N} + BD^{-\alpha_D}$ subject to $6ND = C$ yields the Lagrange stationarity condition:

$$\alpha_N AN^{-\alpha_N} = \alpha_D BD^{-\alpha_D}. \quad (3)$$

Solving for N_{opt} and D_{opt} as functions of C produces the optimal allocation frontiers:

$$\boxed{N_{\text{opt}}(C) \propto C^{\frac{\alpha_D}{\alpha_N + \alpha_D}}, \quad D_{\text{opt}}(C) \propto C^{\frac{\alpha_N}{\alpha_N + \alpha_D}}}. \quad (4)$$

For the empirical parameters reported by Hoffmann et al. [19] ($\alpha_N = 0.34, \alpha_D = 0.28$), the optimal scaling exponents are:

$$N_{\text{opt}} \propto C^{\frac{0.28}{0.34+0.28}} = C^{0.45}, \quad D_{\text{opt}} \propto C^{\frac{0.34}{0.34+0.28}} = C^{0.55}. \quad (5)$$

This establishes the ****Chinchilla Prescription****: model size and training data should scale in roughly equal proportions ($N \propto D$), in stark contrast to earlier Kaplan et al. [24] estimates that favored scaling parameters much faster than tokens ($N \propto C^{0.73}, D \propto C^{0.27}$).

Recent replication and sensitivity studies have clarified this discrepancy. Besiroglu et al. [4] demonstrated statistical sensitivity in the nonlinear regression fits. Porian et al. [32] showed that accounting for non-embedding parameter FLOPs, learning rate warmup schedules, and scale-dependent optimizer tuning fully reconciles the Kaplan and Chinchilla datasets.

2.2 Essential Distinctions: Loss vs. Capabilities and Broken Scaling

Two critical distinctions must be maintained when interpreting scaling laws:

- **Continuous Cross-Entropy Loss vs. Discontinuous Capability Metrics:** While pre-training cross-entropy loss $L(N, D)$ improves smoothly as a power law, downstream accuracy on discrete tasks (e.g., arithmetic, multi-step code execution, translation benchmarks) often appears to jump discontinuously at critical thresholds. Schaeffer, Miranda, and Koyejo [36] proved that these “emergent abilities” [43] are frequently mathematical artifacts of nonlinear, thresholded metrics (such as exact-string match 0/1 accuracy). When evaluated on smooth, continuous metrics (such as token edit distance or average log-probability), the underlying capabilities improve smoothly and predictably with scale.

- **Single Power Laws vs. Broken Power Laws:** Empirical scaling laws are valid over finite measurement windows. Caballero et al. [7] introduced broken-power-law fits, formalizing how curves transition between different scaling regimes, experience plateauing due to data repetition [30], or saturate as they approach the irreducible entropy floor.

3 Information-Theoretic Foundations and Shannon Limits

3.1 Cross-Entropy Decomposition and the Irreducible Bayes Floor

Let $\mathcal{X}^{\mathbb{N}}$ be the space of infinite token sequences drawn from an underlying stationary, ergodic ground-truth language process P^* . Let $Z_t = (X_1, \dots, X_t)$ denote the conditioning context up to step t , and $Q(X_{t+1} | Z_t)$ denote the model’s predictive distribution. The expected cross-entropy loss decomposes exactly into:

$$\mathbb{E}_{P^*}[-\log Q(X_{t+1} | Z_t)] = H_{P^*}(X_{t+1} | Z_t) + \mathbb{E}_{Z_t}[D_{\text{KL}}(P^*(\cdot | Z_t) \| Q(\cdot | Z_t))]. \quad (6)$$

Because the Kullback–Leibler divergence is non-negative and vanishes if and only if $Q \equiv P^*$, the conditional source entropy $H_{P^*}(X_{t+1} | Z_t)$ represents the ****absolute, unbeatable Bayes risk**** for log loss.

For any stationary process on a finite vocabulary $\mathcal{V}$, the conditional entropy $H(X_{t+1} | X_1^t)$ is monotonically non-increasing in t and converges to the source entropy rate [12, 37]:

$$h = \lim_{t \rightarrow \infty} H(X_{t+1} | X_1^t) = \lim_{t \rightarrow \infty} \frac{1}{t} H(X_1, \dots, X_t). \quad (7)$$

Thus, the floor E in empirical scaling laws satisfies $E \geq h$. Claude Shannon’s 1951 human letter-guessing experiment [38] estimated the entropy of English text to be approximately 0.6 to 1.3 bits/character (≈ 0.4 to 0.9 nats/char). Frontier autoregressive models achieve validation cross-entropy losses near 1.0–1.4 nats/token (≈ 0.6 – 0.85 bits/char), approaching this fundamental Shannon limit.

3.2 Why Scaling is Rate–Distortion, Not a Channel Reliability Exponent

A common heuristic intuition attempts to link neural scaling to Shannon’s ****channel coding reliability error exponent****. We formalize why this analogy is structurally flawed and establish that the correct Shannon-theoretic framework is ****Rate–Distortion Theory****.

Conceptual Intuition: Error Exponents vs. Distortion–Rate

Channel Coding Error Exponent: For communication over a noisy channel with block-length n at transmission rate $R < C$, the probability of decoding error decays *exponentially*:

$$P_e(n) \leq e^{-nE_r(R)}.$$

Neural Scaling Law: The excess test loss decays *polynomially* in parameter count N :

$$L(N) - E \sim N^{-\alpha_N}.$$

Why the difference? In channel coding, independent channel noise across n channel uses drives exponential concentration of typical sets. In neural scaling, parameters N provide a compressed representation ($R \approx Nb$ bits) of an infinite-dimensional, continuous or combinatorial data distribution. Excess loss measures the reconstruction distortion $D(R)$, which obeys a power law whenever the source covariance spectrum has a heavy polynomial tail.

Theorem 3.1 (Gaussian Reverse Water-Filling on Power-Law Spectra). Let X be a zero-mean Gaussian random element in a separable Hilbert space with covariance eigenvalues decaying as a power law:

$$\lambda_k = c \cdot k^{-(1+\rho)}, \quad c > 0, \quad \rho > 0, \quad k \in \mathbb{N}^+. \quad (8)$$

Under mean squared error distortion $D = \mathbb{E}[\|X - \hat{X}\|^2]$, the distortion-rate function $D(R)$ in the asymptotic high-rate regime ($R \rightarrow \infty$) satisfies:

$$\boxed{D(R) \asymp R^{-\rho}}. \quad (9)$$

Proof. Under Gaussian rate-distortion theory [3, 12], the optimal rate $R(\theta)$ and distortion $D(\theta)$ are parameterized by the water-filling threshold $\theta > 0$:

$$D(\theta) = \sum_{k=1}^{\infty} \min(\theta, \lambda_k), \quad R(\theta) = \frac{1}{2} \sum_{k=1}^{\infty} \log_2 \max\left(1, \frac{\lambda_k}{\theta}\right). \quad (10)$$

Only modes with $\lambda_k > \theta$ are actively encoded. The number of active modes is:

$$K(\theta) = \max\{k \in \mathbb{N} : ck^{-(1+\rho)} \geq \theta\} = \left\lfloor \left(\frac{c}{\theta}\right)^{\frac{1}{1+\rho}} \right\rfloor \asymp \theta^{-\frac{1}{1+\rho}}. \quad (11)$$

For active modes $k \leq K(\theta)$, we have $\frac{\lambda_k}{\theta} = \left(\frac{K}{k}\right)^{1+\rho}$. The total rate in bits is:

$$\begin{aligned} R(\theta) &= \frac{1+\rho}{2} \sum_{k=1}^{K(\theta)} \log_2 \left(\frac{K}{k}\right) = \frac{1+\rho}{2} \log_2 \left(\frac{K^K}{K!}\right) \\ &= \frac{1+\rho}{2 \ln 2} [K \ln K - (K \ln K - K + \mathcal{O}(\ln K))] \\ &= \frac{(1+\rho) \log_2 e}{2} K(\theta) + \mathcal{O}(\log K) \asymp K(\theta). \end{aligned} \quad (12)$$

Thus, the description rate is directly proportional to the number of active modes: $R \asymp K \asymp \theta^{-\frac{1}{1+\rho}}$, which implies $\theta(R) \asymp R^{-(1+\rho)}$.

The total distortion $D(\theta)$ is the sum of noise injected into active modes plus the unencoded tail variance:

$$\begin{aligned} D(\theta) &= K(\theta)\theta + \sum_{k=K(\theta)+1}^{\infty} ck^{-(1+\rho)} \\ &\approx K(\theta)\theta + \int_{K(\theta)}^{\infty} cx^{-(1+\rho)} dx = K(\theta)\theta + \frac{c}{\rho} K(\theta)^{-\rho}. \end{aligned} \quad (13)$$

Substituting $K(\theta) \asymp (c/\theta)^{\frac{1}{1+\rho}}$ into $D(\theta)$:

$$D(\theta) \asymp \theta \left(\frac{c}{\theta}\right)^{\frac{1}{1+\rho}} + \frac{c}{\rho} \left[\left(\frac{c}{\theta}\right)^{\frac{1}{1+\rho}}\right]^{-\rho} = \left(1 + \frac{1}{\rho}\right) c^{\frac{1}{1+\rho}} \theta^{\frac{\rho}{1+\rho}}. \quad (14)$$

Finally, substituting $\theta(R) \asymp R^{-(1+\rho)}$ gives:

$$D(R) \asymp \left(R^{-(1+\rho)}\right)^{\frac{\rho}{1+\rho}} = R^{-\rho}. \quad (15)$$

□

Corollary 3.2 (Finite-Dimensional vs. Heavy-Tailed Spectra). If the source is strictly d -dimensional ($\lambda_k = 0$ for $k > d$), all d modes are active at high rate, yielding exponential distortion decay:

$$D(R) \asymp 2^{-2R/d} = \exp\left(-\frac{2 \ln 2}{d} R\right). \quad (16)$$

Foundational Takeaway: The Spectral Fingerprint

Power-law scaling $L(N) \sim N^{-\alpha}$ is the **mathematical fingerprint** of an effectively infinite-dimensional, heavy-tailed data spectrum ($\lambda_k \sim k^{-(1+\rho)}$). If real data were finite-dimensional, optimal compression would yield exponential error decay.

3.3 Metric Entropy and Yang–Barron Minimax Estimation

Yang and Barron (1999) [44] established the definitive information-theoretic bounds for statistical estimation by linking minimax risk to the Kolmogorov ε -covering entropy $H(\varepsilon; \mathcal{P}) = \log \mathcal{N}(\varepsilon, \mathcal{P}, \rho)$.

Theorem 3.3 (Yang–Barron Minimax Balance [42, 44]). Let $\mathcal{P}$ be a family of probability distributions on $\mathcal{X}$. If the metric entropy under Hellinger distance satisfies $H(\varepsilon; \mathcal{P}) \asymp \varepsilon^{-r}$ for some $r > 0$, then the minimax estimation risk over n i.i.d. samples satisfies:

$$\inf_{\hat{p}_n} \sup_{p^* \in \mathcal{P}} \mathbb{E}_{p^*} [D_{\text{KL}}(p^* \| \hat{p}_n)] \asymp \varepsilon_n^2, \quad (17)$$

where the minimax optimal rate ε_n is the unique solution to the balance equation:

$$\boxed{H(\varepsilon_n; \mathcal{P}) \asymp n \varepsilon_n^2}. \quad (18)$$

Corollary 3.4 (Sobolev Nonparametric Rates). For target densities in a Sobolev ball $W^s(\mathbb{R}^d)$ of smoothness s in d dimensions, classical metric entropy gives $H(\varepsilon) \asymp \varepsilon^{-d/s}$. Solving $\varepsilon_n^{-d/s} \asymp n \varepsilon_n^2$ gives $\varepsilon_n^{2+d/s} \asymp n^{-1}$, yielding the minimax risk rate:

$$\mathcal{R}_n \asymp n^{-\frac{2s}{2s+d}} \implies \alpha_D = \frac{2s}{2s+d}. \quad (19)$$

3.4 Information-Theoretic Model/Data Allocation: Jeon & Van Roy (2024)

Jeon and Van Roy [22] formalized neural scaling through a Bayesian rate–distortion decomposition. For a dataset $\mathcal{D}_T$ of T observations and a target data-generating process F approximated by a compressed surrogate model $\tilde{F}$, the expected excess log loss decomposes as:

$$\mathcal{E}_T \lesssim \underbrace{\frac{I(F; \tilde{F})}{T}}_{\text{Estimation / Information Cost}} + \underbrace{\mathbb{E} D_{\text{KL}}(P_F(\cdot | X) \| P_{\tilde{F}}(\cdot | X))}_{\text{Representation Misspecification Cost}}. \quad (20)$$

The first term is the statistical price of identifying $\tilde{F}$ from T tokens, which decreases as T increases. The second term is the structural approximation error, which decreases as surrogate capacity N increases. For an infinite-width two-layer network prior, optimizing this bound under a compute constraint $C \approx NT$ rigorously yields an optimal data-to-model allocation $N \propto T$ up to logarithmic factors, providing an information-theoretic foundation for the Chinchilla scaling rule.

4 Statistical Physics of Language and Horizon-Limited Scaling

4.1 Hilberg’s Law and Long-Range Context Correlations

In 1990, Wolfgang Hilberg [18] conjectured that natural language possesses long-range power-law memory. Specifically, the block entropy of n consecutive words or characters scales as:

$$H(X_1^n) = hn + cn^\delta + o(n^\delta), \quad 0 < \delta < 1 \quad (\text{typically } \delta \approx 0.5). \quad (21)$$

Differentiating or differencing Equation (21) implies that the excess conditional entropy of the $(n+1)$ -th token given context length n decays as a power law [14]:

$$H(X_{n+1} | X_1^n) - h \asymp n^{-\gamma}, \quad \text{where } \gamma = 1 - \delta. \quad (22)$$

4.2 Derivation of the Data Scaling Exponent: Cagnetta et al. (2024/2026)

A milestone breakthrough by Cagnetta, Raventós, Ganguli, and Wyart [9] established an analytical link between measurable language statistics and the empirical data scaling exponent α_D .

Conceptual Intuition: Reading with Noisy Glasses

Imagine learning to predict the next word in a language by reading D total tokens. 1. Long-range grammatical and semantic correlations decay with distance as $|C(n)| \sim n^{-\beta}$. 2. Estimating these correlations from a finite dataset of D tokens is corrupted by central limit sampling noise $\sigma_{\text{sample}} \sim \mathcal{O}(D^{-1/2})$. 3. The model can only exploit correlations that rise above the noise floor: $n^{-\beta} \gtrsim D^{-1/2}$. This sets a maximum **statistically resolvable context horizon**:

$$n^*(D) \sim D^{\frac{1}{2\beta}}.$$

4. If the model is expressive enough to master all patterns within this horizon, the remaining unpredicted entropy is governed by the conditional entropy gap at horizon n^* :

$$\mathcal{L}(D) - H_\infty \sim (n^*(D))^{-\gamma} = \left(D^{\frac{1}{2\beta}}\right)^{-\gamma} = D^{-\frac{\gamma}{2\beta}}.$$

Theorem 4.1 (Cagnetta–Raventós–Ganguli–Wyart Scaling Law [9]). Suppose a language corpus satisfies:

1. **Conditional Entropy Decay:** $H_n - H_\infty \asymp n^{-\gamma}$, where $H_n = H(X_{t+1} | X_{t-n+1}^t)$.
2. **Token Covariance Decay:** $\|\Sigma(n)\|_{\text{op}} \asymp n^{-\beta}$, where $\Sigma(n)$ is the token correlation matrix at lag n .
3. **Fast Within-Horizon Learning:** Within the resolvable horizon $n \leq n^*(D)$, the model learns patterns sufficiently fast that the dominant bottleneck is context truncation at $n^*(D)$.

Then, when trained on a dataset of D tokens in the data-limited regime, the excess cross-entropy loss obeys:

$$\boxed{\mathcal{L}(D) - H_\infty \asymp D^{-\alpha_D}, \quad \text{where } \alpha_D = \frac{\gamma}{2\beta}.} \quad (23)$$

Empirical Validation without Parameter Fitting. Cagnetta et al. measured γ and β directly on language corpora and compared the predicted $\alpha_D = \gamma/(2\beta)$ against actual training curves of GPT-2 and LLaMA transformers:

$$\begin{aligned} \textbf{TinyStories: } \quad & \gamma = 0.325 \pm 0.003, \quad \beta = 0.88 \pm 0.06 \\ & \implies \alpha_D^{\text{pred}} = 0.185 \pm 0.013 \quad (\alpha_D^{\text{emp}} \approx 0.18), \end{aligned} \quad (24)$$

$$\begin{aligned} \textbf{WikiText-103: } \quad & \gamma = 0.265 \pm 0.016, \quad \beta = 0.94 \pm 0.16 \\ & \implies \alpha_D^{\text{pred}} = 0.141 \pm 0.025 \quad (\alpha_D^{\text{emp}} \approx 0.14). \end{aligned} \quad (25)$$

The empirical scaling curves collapse onto a universal curve under the theoretical rescaling $D \mapsto D/n^{2\beta}$ and $\mathcal{L}_n \mapsto n^\gamma \mathcal{L}_n$. This represents the first parameter-free, first-principles prediction of a neural scaling exponent from measurable corpus statistics.

4.3 Controlled Data Structure: Random Walk Graphs (Barkeshli et al., 2026)

Barkeshli, Alfarano, and Gromov [2] trained transformers on synthetic language generated by random walks on graphs with tunable topology. They found that even for Erdős–Rényi graphs lacking heavy-tailed power-law spectra, neural networks exhibit power-law scaling over finite ranges. This demonstrates that while heavy-tailed language statistics analytically set exponents in the horizon-limited regime, neural optimization dynamics can also generate effective power-law scaling over finite windows.

5 High-Dimensional Geometry, Manifold Tiling, and Approximation Limits

5.1 Manifold Tiling Geometry: Sharma & Kaplan (2020)

High-dimensional data in $\mathbb{R}^{D_{\text{amb}}}$ typically concentrates on a lower-dimensional Riemannian submanifold $\mathcal{M}$ of intrinsic dimension $d \ll D_{\text{amb}}$. Sharma and Kaplan [39] derived the parameter exponent α_N by modeling neural networks as piecewise-affine approximators on $\mathcal{M}$.

Geometric Intuition: Tiling a Curved Manifold with Flat Tiles

Consider approximating a curved 2D surface ($d = 2$) in 3D space using N flat triangular tiles: 1. Each tile has a characteristic linear diameter $\varepsilon \asymp N^{-1/d} = N^{-1/2}$. 2. A smooth $\mathcal{C}^2$ target function f^* curves away from each flat affine tile $\hat{f}$ by its second-order Taylor remainder:

$$|f^*(x) - \hat{f}(x)| \leq \frac{1}{2} \|x - x_0\|^2 \|\nabla^2 f^*(x_0)\| \sim \varepsilon^2 = N^{-2/d}.$$

3. Integrating the squared error over all N tiles yields the mean squared loss:

$$L(N) = \int_{\mathcal{M}} |f^*(x) - \hat{f}(x)|^2 d\mu(x) \asymp \varepsilon^4 = (N^{-1/d})^4 = N^{-4/d}.$$

Theorem 5.1 (Manifold Tiling Exponent [39]). Let $\mathcal{M}$ be a compact d -dimensional Riemannian manifold and $f^* \in \mathcal{C}^2(\mathcal{M})$. If a neural network with N parameters partitions $\mathcal{M}$ into $K \asymp N$ local linear regions of size $\varepsilon \asymp N^{-1/d}$, then under squared error regression:

$$L(N) - E \asymp N^{-4/d} \implies \alpha_N = \frac{4}{d}. \quad (26)$$

If the target function is only Lipschitz continuous ($\mathcal{C}^1$), the first-order error yields $\alpha_N = 2/d$.

Intrinsic Dimension Measurements. For GPT models where empirical parameter scaling gives $\alpha_N \approx 0.076$, the geometric formula predicts an effective intrinsic dimension $d \approx 4/\alpha_N \approx 53$. Sharma and Kaplan measured intrinsic dimensions of transformer token activations and found $d \approx 7$ within a single contiguous 1024-token document, but $d > 90$ across heterogeneous documents, revealing a scale-dependent multi-scale manifold hierarchy.

5.2 Resolution-Limited vs. Variance-Limited Regimes: Bahri et al. (PNAS 2024)

Bahri, Dyer, Kaplan, Lee, and Sharma [1] established a comprehensive phase diagram distinguishing variance bottlenecks from resolution bottlenecks along both model and data axes:

Scaling Axis	Limiting Mechanism	Asymptotic Error Scaling
Data (<i>D</i> -Variance)	Sample variance near infinite-data limit	$\Delta L(D) \propto D^{-1}$ (Central Limit Theorem)
Width (<i>w</i> -Variance)	Ensemble variance near infinite-width limit	$\Delta L(w) \propto w^{-1}$ (Gaussian process convergence)
Data (<i>D</i> -Resolution)	Finite sample coverage of <i>d</i> -manifold	$\Delta L(D) \propto D^{-4/d}$ (or $D^{-2/d}$)
Model (<i>N</i> -Resolution)	Parameter cell coverage of <i>d</i> -manifold	$\Delta L(N) \propto N^{-4/d}$ (or $N^{-2/d}$)

Table 2: The four fundamental scaling regimes of Bahri et al. [1].

The transition between these regimes is accurately described by a smooth harmonic crossover:

$$L(N, D) = E + \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\kappa}} + \left(\frac{D_c}{D} \right)^{\frac{\alpha_D}{\kappa}} \right]^\kappa, \quad \kappa > 0. \quad (27)$$

5.3 Fundamental Impossibility Lower Bounds in Approximation Theory

Theoretical scaling laws are bounded by foundational impossibility theorems in approximation theory, information theory, and computational complexity:

Theorem 5.2 (DeVore–Howard–Micchelli Continuous Non-Linear *n*-Width Lower Bound (1989) [15]). Let $W_p^s([0, 1]^d)$ be the Sobolev ball of smoothness *s* in *d* dimensions. Let $\mathcal{M}_N = \{g(\theta, \cdot) : \theta \in \mathbb{R}^N\}$ be any parametric family where parameters $\theta(f)$ are selected continuously from *f*. Then the continuous non-linear *n*-width satisfies:

$$d_N(W_p^s([0, 1]^d))_q \geq c \cdot N^{-s/d}. \quad (28)$$

This theorem proves that **no continuous parameter optimization method** (including gradient descent on neural networks) can ever circumvent the curse of dimensionality $N^{-s/d}$ for worst-case Sobolev functions.

Theorem 5.3 (Rissanen MDL Parametric Redundancy Bound (1986, 1996) [33, 34]). For any regular, identifiable *k*-dimensional parametric family $\{p_\theta : \theta \in \Theta \subset \mathbb{R}^k\}$, universal coding incurs an unavoidable minimax parameter redundancy:

$$\inf_q \sup_{\theta \in \Theta} \mathbb{E}_{p_\theta} \left[\log \frac{p_\theta(X^n)}{q(X^n)} \right] \geq \frac{k}{2} \log n + \frac{k}{2} \log \left(\frac{1}{2\pi e} \right) + \log \int_\Theta \sqrt{\det I(\theta)} d\theta + o(1). \quad (29)$$

Theorem 5.4 (Statistical Query (SQ) Lower Bounds (Kearns 1998, Shamir 2020) [25]). For parity functions on d binary bits, any algorithm with Statistical Query access with tolerance $\tau \geq 2^{-d/3}$ requires at least $2^{\Omega(d)}$ queries. Furthermore, under cryptographic assumptions, there exist functions that no polynomial-size neural network can learn with loss advantage $> 2^{-\Omega(d)}$.

Takeaway: Scaling Laws are Data-Contingent

Neural scaling laws $L \sim N^{-\alpha}$ are ****not** universal mathematical laws of all conceivable functions**. They reflect the structural properties of natural multimodal data: low intrinsic manifold dimensionality, heavy-tailed spectral decay, and compositional hierarchy.

6 Statistical Physics, Kernel Theory, and Solvable Models

6.1 Kernel Spectra and the Hard-Cutoff Heuristic

In the infinite-width Neural Tangent Kernel (NTK) regime [11, 21], neural network learning curves can be solved analytically. Let T_K be the kernel integral operator with eigenvalues $\lambda_k \asymp k^{-a}$ ($a > 1$) and target function projections $c_k^2 \asymp k^{-b}$ ($b > 1$).

Bordelon, Canatar, and Pehlevan [6, 10] established that neural networks exhibit ****spectral bias****: high-eigenvalue (low-frequency) modes are learned first. Suppose a learning algorithm resolves all modes up to an effective spectral cutoff k_* and leaves higher modes unlearned. The residual error is the unlearned target power:

$$\mathcal{E} \approx \sum_{k > k_*} c_k^2 \asymp \int_{k_*}^{\infty} x^{-b} dx \asymp k_*^{-(b-1)}. \quad (30)$$

For ****ridgeless kernel regression**** on n samples, the effective regularization parameter κ solves $n \asymp \sum_k \frac{\lambda_k}{\lambda_k + \kappa} \asymp \kappa^{-1/a} \implies \kappa \asymp n^{-a}$. The cutoff mode satisfies $\lambda_{k_*} \asymp \kappa \implies k_*^{-a} \asymp n^{-a} \implies k_* \asymp n$. Substituting $k_* \asymp n$ into Equation (30) gives:

$$\mathcal{E}(n) \asymp n^{-(b-1)}. \quad (31)$$

Under noisy labels with optimal regularization, Cui et al. [13] show the exponent transitions to $n^{-(b-1)/b}$.

6.2 Exact Joint (N, D) Scaling Laws in Linear Regression (Lin et al., NeurIPS 2024)

Lin, Wu, Kakade, Bartlett, and Lee [27] (see also Lin, Dobriban, Hong [26]) derived exact non-asymptotic and asymptotic joint scaling laws for high-dimensional linear regression trained by one-pass SGD with a random feature sketch of size N on D samples.

Theorem 6.1 (Lin et al. (NeurIPS 2024) [27]). Let covariate covariance eigenvalues decay as $\lambda_k \asymp k^{-a}$ ($a > 1$) and target coefficients decay as $c_k^2 \asymp k^{-b}$ ($1 < b < a+1$). Under one-pass SGD with feature sketch dimension N and sample count D , the excess generalization error satisfies:

$$\mathcal{E}(N, D) = \Theta \left(N^{-(b-1)} + D^{-\frac{b-1}{a}} \right). \quad (32)$$

Under an isotropic parameter prior ($b = a$), this simplifies to $\Theta(N^{-(a-1)} + D^{-\frac{a-1}{a}})$.

Intuition: Why Do the Model and Data Exponents Differ?

Both terms in Equation (32) originate from the cutoff formula $\mathcal{E} \asymp k_*^{-(b-1)}$:

- ****Parameter Cutoff:**** Projecting into an N -dimensional sketch truncates modes at index $k_* = N$. The residual error is $k_*^{-(b-1)} = N^{-(b-1)}$.
- ****One-Pass SGD Data Cutoff:**** One-pass SGD does not invert the full covariance matrix. A mode k is learned only when accumulated gradient signal over D steps reaches order one: $D\lambda_k \gtrsim 1 \implies Dk^{-a} \gtrsim 1 \implies k_* \asymp D^{1/a}$. The residual error is $k_*^{-(b-1)} = (D^{1/a})^{-(b-1)} = D^{-\frac{b-1}{a}}$.

Key Prediction: $\alpha_N = b - 1$ and $\alpha_D = (b - 1)/a \implies \alpha_N = a \cdot \alpha_D$. For Chinchilla empirical values ($\alpha_N = 0.34, \alpha_D = 0.28$), this implies $a \approx 1.21$ and $b \approx 1.34$, perfectly within the theoretical admissible window.

6.3 Dynamical Training Phases: Maloney et al., Bordelon et al., Paquette et al.

Maloney, Roberts, and Sully [28] provided a solvable random-feature model demonstrating distinct compute-optimal phases. Bordelon, Atanasov, and Pehlevan [5] extended this to gradient descent dynamics, explaining why learning curve exponents differ across early-time and late-time width limits. Paquette et al. [31] rigorously proved the existence of $4 + 3$ distinct compute-optimal scaling phases depending on the relative complexity of data spectrum, target smoothness, and batch size.

7 Discrete Tasks, Associative Memory, and Superposition

7.1 Associative Memories with Heavy-Tailed Inputs (Cabannes et al., 2024)

Cabannes, Dohmatob, and Bietti [8] analyzed high-dimensional outer-product associative memories storing discrete associations drawn from Zipf-distributed inputs. They derived exact sample and parameter scaling curves for key-value associative memories, rigorously bridging discrete memory storage with neural scaling laws.

7.2 The Quantization Model of Neural Scaling (Michaud et al., NeurIPS 2023)

Michaud, Liu, Girit, and Tegmark [29] proposed that knowledge consists of a large discrete set of independent sub-problems, facts, or skills (“quanta”) $\mathcal{Q} = \{q_1, q_2, \dots\}$.

Theorem 7.1 (Quantization Scaling Exponent [29]). Let each quantum q_i have utility $U_i \asymp i^{-q}$ with $q > 1$ (Zipf’s law for skills). Under greedy parameter allocation, a network with N parameters masters the top $K \asymp N$ most useful quanta. The residual loss from unlearned quanta is:

$$L(N) - E = \sum_{i=N+1}^{\infty} U_i \approx \int_N^{\infty} x^{-q} dx = \frac{1}{q-1} N^{-(q-1)} \implies \boxed{\alpha_N = q - 1}. \quad (33)$$

7.3 Hutter’s Countable-Input Memorization Model (2021)

Marcus Hutter [20] analyzed an optimal memorizer on countably many discrete inputs. If input probabilities follow a power law $\theta_i \propto i^{-(1+\eta)}$ ($\eta > 0$), a training set of n samples observes all

inputs with frequency $\theta_i \gtrsim 1/n \implies i_* \asymp n^{\frac{1}{1+\eta}}$. The expected test error over unseen inputs scales as:

$$\mathbb{E}[E_n] = \sum_{i>i_*} \theta_i \asymp i_*^{-\eta} = \left(n^{\frac{1}{1+\eta}}\right)^{-\eta} = \boxed{n^{-\frac{\eta}{1+\eta}}}. \quad (34)$$

7.4 Toy Models of Superposition (Elhage et al., Anthropic 2022)

How does a neural network with activation width d_{model} represent $M \gg d_{\text{model}}$ semantic features? Elhage et al. [16] formalized **neural superposition**:

Theorem 7.2 (Johnson–Lindenstrauss Superposition and Interference Noise). Let M sparse, independent features have active probability $p \ll 1$ and unit variance. An overcomplete dictionary $\mathbf{W} \in \mathbb{R}^{d_{\text{model}} \times M}$ with unit-norm columns satisfies $|\langle w_i, w_j \rangle| \leq \varepsilon$ in dimension $d_{\text{model}} = \mathcal{O}\left(\frac{\log M}{\varepsilon^2}\right)$ by the Johnson–Lindenstrauss lemma [23].

Linear reconstruction $\hat{x}_i = w_i^T(\mathbf{W}x) = x_i + \sum_{j \neq i} \langle w_i, w_j \rangle x_j$ suffers cross-talk interference variance:

$$\mathbb{E} \left[\left(\sum_{j \neq i} \langle w_i, w_j \rangle x_j \right)^2 \right] = p(1-p) \sum_{j \neq i} |\langle w_i, w_j \rangle|^2 \approx \boxed{p(1-p) \frac{M}{d_{\text{model}}}}. \quad (35)$$

As hidden width d_{model} increases, interference noise decays as $\mathcal{O}(1/d_{\text{model}})$, providing a representation capacity mechanism for power-law error reduction.

8 Comparative Synthesis and Taxonomy of Theories

Table 3 provides a unified comparative synthesis of the nine major theoretical scaling frameworks.

The Grand Unifying Pattern: Tail Ranking and Resolution Rules

Read across the taxonomy, a single unifying mathematical structure emerges:

$$\text{Scaling Exponent} = (\text{Tail Decay of Ranked Primitives}) \times (\text{Algorithmic Resolution Rule})$$

Whether ranking covariance eigenvalues $\lambda_k \sim k^{-(1+\rho)}$, kernel modes $c_k^2 \sim k^{-b}$, manifold cells $\varepsilon \sim N^{-1/d}$, Zipfian skills $U_i \sim i^{-q}$, or token correlations $|C(n)| \sim n^{-\beta}$, neural scaling curves arise whenever an algorithm progressively resolves a heavy-tailed hierarchy of features.

9 The “Honest Gap” and Open Research Frontiers

Despite major theoretical breakthroughs, a fundamental epistemic gap remains between theory and practice:

Framework	Core Assumptions	Mathematical Machinery	Derived Exponent	Primary References
Rate–Distortion Theory	Gaussian/heavy-tailed source, spectrum $\lambda_k \sim k^{-(1+\rho)}$	Shannon Reverse Water-Filling, high-rate asymptotics	$D(R) \sim R^{-\rho}$	Shannon (1948) [37], Berger (1971) [3]
Metric Entropy / Minimax	Sobolev/Besov target class $W^s(\mathbb{R}^d)$, metric entropy $H(\varepsilon) \sim \varepsilon^{-d/s}$	Yang–Barron balance $H(\varepsilon_n) \asymp n\varepsilon_n^2$, Fano testing	$\alpha_D = \frac{2s}{2s+d}$	Yang & Barron (1999) [44], Tsybakov (2009) [42]
Language Context Horizon	Correlation decay $ C(n) \sim n^{-\beta}$, conditional entropy $H_n - h \sim n^{-\gamma}$	Sample noise cutoff $n^*(D) \sim D^{1/(2\beta)}$, entropy truncation	$\alpha_D = \frac{\gamma}{2\beta}$	Cagnetta et al. (2024) [9], Hilberg (1990) [18]
Manifold Tiling Geometry	Data on d -dimensional Riemannian manifold $\mathcal{M}$, $\mathcal{C}^2$ target function	Voronoi partition, 2nd-order Taylor remainder on tiles	$\alpha_N = \frac{4}{d}$	Sharma & Kaplan (2020) [39]
Resolution vs. Variance Phase Diagram	Interplay of finite parameter count N and dataset size D	Nonparametric manifold covering + CLT variance	$\Delta L \sim N^{-4/d} + D^{-1}$	Bahri et al. (PNAS 2024) [1]
Kernel Spectra / NTK Physics	Power-law kernel eigenvalues $\lambda_k \sim k^{-a}$, target projections $c_k^2 \sim k^{-b}$	Replica method, Random Matrix Theory, spectral bias	$\mathcal{E}(D) \sim D^{-\frac{a+b-1}{a}}$	Bordelon et al. (2020) [6], Cui et al. (2021) [13]
Linear Model Joint (N, D) Laws	Covariance $\lambda_k \sim k^{-a}$, target decay $c_k^2 \sim k^{-b}$, one-pass SGD	Sketch dimension truncation + accumulated gradient SNR	$\mathcal{E} = \Theta(N^{-(b-1)} + D^{-\frac{b-1}{a}})$	Lin et al. (NeurIPS 2024) [26, 27]
Skill Quantization	Independent skills/quanta with Zipfian utilities $U_i \sim i^{-q}$	Greedy capacity allocation, tail integration of unlearned mass	$\alpha_N = q - 1$	Michaud et al. (NeurIPS 2023) [29]
Feature Superposition	$M \gg d_{\text{model}}$ sparse features, almost-orthogonal frame	Johnson–Lindenstrauss lemma, cross-talk variance	Noise $\propto \frac{pM}{d_{\text{model}}}$	Elhage et al. (2022) [16]

Table 3: Comprehensive master taxonomy of theoretical frameworks for neural scaling laws.

The Fundamental “Honest Gap”

No existing theory derives the empirical scaling exponents ($\alpha_N \approx 0.34, \alpha_D \approx 0.28$) for real natural language purely from first physical principles. Existing theories rigorously prove that scaling exponents are functions of underlying data statistics:

$$\alpha = \Phi(a_{\text{spectral}}, d_{\text{manifold}}, q_{\text{Zipf}}, \gamma_{\text{entropy}}, \beta_{\text{correlation}}). \quad (36)$$

However, these parameters (a, d, q, γ, β) are ****empirically measured rather than derived from axioms of human cognition or linguistics****. Deriving why human language generates these specific spectral exponents represents the ultimate frontier of deep learning theory.

9.1 Key Open Frontiers

1. **Beating Power Laws via Data Pruning:** Sorscher, Geirhos, Shekhar, Ganguli, and Morcos [41] proved that optimizing data quality via metric-based pruning can break polynomial power laws, transitioning learning curves to ****exponential scaling**** ($\mathcal{E} \sim e^{-\alpha D}$). Developing an operational theory of “effective data quality” that incorporates deduplication and curriculum difficulty remains a high-priority challenge.
2. **Test-Time Compute and Search Scaling:** Recent models (e.g., OpenAI o1/o3, Snell et al. [40]) demonstrate that spending compute at inference time (via tree search, chain-of-thought verification, and self-correction) yields scaling gains that can substitute for pre-training scale. A unified rate–distortion theory bridging pretraining parameter rate with test-time search budget is an active frontier.
3. **Feature Learning Beyond Fixed Kernels:** Extending kernel and NTK scaling theories beyond the “lazy training” regime [11] to capture representation learning, self-attention routing, and rotary positional embeddings.

10 Conclusion

Neural scaling laws are not empirical accidents; they are the rigorous macroscopic manifestation of scale-free mathematical structures.

By analyzing scaling through Shannon rate–distortion theory, we establish that power laws reflect lossy compression of heavy-tailed, infinite-dimensional data spectra under reverse water-filling. Through high-dimensional geometry and statistical physics, parameter and data exponents map directly to manifold dimensionality, kernel eigenspectra, and long-range correlation decay. While fundamental lower bounds (Yang–Barron, DeVore–Howard–Micchelli, Rissanen) establish unbeatable speed limits on learning, the rich hierarchical structure of natural language permits modern deep neural networks to operate along an optimal information-theoretic frontier.

References

- [1] Yasaman Bahri, Ethan Dyer, Jared Kaplan, Jaehoon Lee, and Utkarsh Sharma. Explaining Neural Scaling Laws. *Proceedings of the National Academy of Sciences (PNAS)*, 121(27):e2311878121, 2024.
- [2] Maissam Barkeshli, Alberto Alfarano, and Andrey Gromov. On the Origin of Neural Scaling Laws: From Random Graphs to Natural Language. *arXiv preprint arXiv:2601.10684*, 2026.

- [3] Toby Berger. *Rate Distortion Theory: A Mathematical Basis for Data Compression*. Prentice-Hall, 1971.
- [4] Tamay Besiroglu, Ege Erdil, Matthew Barnett, and Josh You. Chinchilla Scaling: A Replication Attempt. *arXiv preprint arXiv:2404.10102*, 2024.
- [5] Blake Bordelon, Alexander Atanasov, and Cengiz Pehlevan. A Dynamical Model of Neural Scaling Laws. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 4345–4382, 2024.
- [6] Blake Bordelon, Abdulkadir Canatar, and Cengiz Pehlevan. Spectrum Dependent Learning Curves in Kernel Regression and Wide Neural Networks. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *Proceedings of Machine Learning Research*, pages 1024–1034, 2020.
- [7] Ethan Caballero, Kshitij Gupta, Irina Rish, and David Krueger. Broken Neural Scaling Laws. In *International Conference on Learning Representations (ICLR)*, 2023.
- [8] Vivien Cabannes, Elvis Dohmatob, and Alberto Bietti. Scaling Laws for Associative Memories. In *International Conference on Learning Representations (ICLR)*, 2024.
- [9] Francesco Cagnetta, Bernat Raventós, Surya Ganguli, and Matthieu Wyart. Deriving Neural Scaling Laws from the Statistics of Natural Language. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2406.01234.
- [10] Abdulkadir Canatar, Blake Bordelon, and Cengiz Pehlevan. Spectral Bias and Task-Model Alignment Explain Generalization in Kernel Regression and Infinitely Wide Neural Networks. *Nature Communications*, 12:2914, 2021.
- [11] Lénaïc Chizat, Edouard Oyallon, and Francis Bach. On Lazy Training in Differentiable Programming. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- [12] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2 edition, 2006.
- [13] Hugo Cui, Bruno Loureiro, Florent Krzakala, and Lenka Zdeborová. Generalization Error Rates in Kernel Regression: The Crossover from the Noiseless to Noisy Regime. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 10131–10143, 2021.
- [14] Łukasz Dębowski. The Relaxed Hilberg Conjecture: A Review and New Experimental Support. *Journal of Quantitative Linguistics*, 22(4):311–337, 2015.
- [15] Ronald A. DeVore, Ralph Howard, and Charles Micchelli. Optimal Nonlinear Approximation. *Manuscripta Mathematica*, 63(4):469–478, 1989.
- [16] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy Models of Superposition. Transformer Circuits Thread, 2022.
- [17] Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md. Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. Deep Learning Scaling Is Predictable, Empirically. *arXiv preprint arXiv:1712.00409*, 2017.
- [18] Wolfgang Hilberg. Der bekannte Grenzwert der redundanzfreien Information in Texten—eine Fehlinterpretation der Shannonschen Experimente? *Frequenz*, 44(9–10):243–248, 1990.

- [19] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training Compute-Optimal Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 30016–30030, 2022.
- [20] Marcus Hutter. Learning Curve Theory. *arXiv preprint arXiv:2102.04074*, 2021.
- [21] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural Tangent Kernel: Convergence and Generalization in Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pages 8571–8580, 2018.
- [22] Hong Jun Jeon and Benjamin Van Roy. Information-Theoretic Foundations for Neural Scaling Laws. *arXiv preprint arXiv:2402.04217*, 2024.
- [23] William B. Johnson and Joram Lindenstrauss. Extensions of Lipschitz Mappings into a Hilbert Space. In *Conference in Modern Analysis and Probability*, volume 26 of *Contemporary Mathematics*, pages 189–206. American Mathematical Society, 1984.
- [24] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling Laws for Neural Language Models. *arXiv preprint arXiv:2001.08361*, 2020.
- [25] Michael Kearns. Efficient Noise-Tolerant Learning from Statistical Queries. *Journal of the ACM*, 45(6):983–1006, 1998.
- [26] Licong Lin, Edgar Dobriban, and Mingyi Hong. Scaling Laws in Linear Regression: Non-Asymptotic and Asymptotic Analysis. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [27] Licong Lin, Jingfeng Wu, Sham M. Kakade, Peter L. Bartlett, and Jason D. Lee. Scaling Laws in Linear Regression: Compute, Parameters, and Data. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [28] Alexander Maloney, Daniel A. Roberts, and James Sully. A Solvable Model of Neural Scaling Laws. *arXiv preprint arXiv:2210.16859*, 2022.
- [29] Eric J. Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The Quantization Model of Neural Scaling. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [30] Niklas Muennighoff, Alexander M. Rush, Boaz Barak, Teven Le Scao, Aleksandra Piktus, Nouamane Tazi, Sampo Pyysalo, Thomas Wolf, and Colin Raffel. Scaling Data-Constrained Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [31] Elliot Paquette, Courtney Paquette, Lechao Xiao, and Jeffrey Pennington. 4+3 Phases of Compute-Optimal Neural Scaling Laws. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [32] Tomer Porian, Mitchell Wortsman, Jenia Jitsev, Ludwig Schmidt, and Yair Carmon. Resolving Discrepancies in Compute-Optimal Scaling of Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, 2024.
- [33] Jorma Rissanen. Stochastic Complexity and Modeling. *The Annals of Statistics*, 14(3):1080–1100, 1986.

- [34] Jorma Rissanen. Fisher Information and Stochastic Complexity. *IEEE Transactions on Information Theory*, 42(1):40–47, 1996.
- [35] Jonathan S. Rosenfeld, Amir Rosenfeld, Yonatan Belinkov, and Nir Shavit. A Constructive Prediction of the Generalization Error Across Scales. In *International Conference on Learning Representations (ICLR)*, 2020.
- [36] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are Emergent Abilities of Large Language Models a Mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [37] Claude E. Shannon. A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3–4):379–423, 623–656, 1948.
- [38] Claude E. Shannon. Prediction and Entropy of Printed English. *Bell System Technical Journal*, 30(1):50–64, 1951.
- [39] Utkarsh Sharma and Jared Kaplan. A Neural Scaling Law from the Dimension of the Data Manifold. *arXiv preprint arXiv:2004.10802*, 2020.
- [40] Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM Test-Time Compute Optimally Can Be More Effective than Scaling Model Parameters. In *International Conference on Learning Representations (ICLR)*, 2025.
- [41] Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos. Beyond Neural Scaling Laws: Beating Power Law Scaling via Data Pruning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 19523–19536, 2022.
- [42] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer, 2009.
- [43] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, Ed H. Chi, Tatsunori Hashimoto, Oriol Vinyals, Percy Liang, Jeff Dean, and William Fedus. Emergent Abilities of Large Language Models. *Transactions on Machine Learning Research (TMLR)*, 2022.
- [44] Yuhong Yang and Andrew Barron. Information-Theoretic Determination of Minimax Rates of Convergence. *The Annals of Statistics*, 27(5):1564–1599, 1999.